



Aviation Investigation Final Report

Location:	Bluestem, Washington	Accident Number:	WPR24FA200
Date & Time:	June 19, 2024, 16:20 Local	Registration:	N323TT
Aircraft:	ROBINSON HELICOPTER COMPANY R44 II	Aircraft Damage:	Destroyed
Defining Event:	Unknown or undetermined	Injuries:	2 Fatal
Flight Conducted Under:	Part 91: General aviation - Personal		

Analysis

The pilot and pilot-rated passenger were returning the helicopter to its home airport after it had undergone routine maintenance, which included the installation of a fuel flow transducer. While in day visual meteorological conditions (VMC), in cruise flight, and about 745 ft above ground level (agl), the helicopter entered a descent. During the descent, the main rotor blades impacted the tailcone, severing the empennage, while about the same time the main rotor's pitch horns and control rod ends fractured and separated. The lack of tail rotor and main rotor components resulted in a total loss of helicopter control. The helicopter impacted terrain at a high descent rate in a nose-high attitude as evidenced by the flattening and upward bending of the tailcone and the fracturing of the engine case. A postaccident fire destroyed most of the wreckage.

Examination of the wreckage revealed evidence of a total loss of engine power as indicated by the lack of rotational scoring on the upper sheave and the undamaged section of the engine cooling fan. Main rotor blade impact with the tailcone and damage to the main rotor system were indicative of main rotor blade blow back and stall due to low rotor rpm. Examination of the freewheeling unit (sprag clutch) revealed normal operation. Multiple B-nuts on the fuel lines associated with the transducer and fuel servo were found loose; however, due to the thermal exposure, it could not be determined if the loss of torque to the B-nuts was due to heat damage or other factors.

The pilot was a highly experienced airplane pilot but had only accumulated about 350 hours of flight experience in helicopters. Based on the helicopter's altitude and the wreckage signatures, after the loss of total power the pilot likely did not reflexively and simultaneously lower the collective, apply aft cyclic, and trim the helicopter to enter an autorotation while in

cruise flight, and/or improperly managed the rotor rpm during the remainder of the autorotation. The lack of timely action allowed the main rotor rpm to reduce to an unrecoverable speed and likely resulted in the main rotor blades stalling, as detailed in the manufacturer’s Safety Notices and FAA guidance documents. The reason for the pilot’s failure to manage rotor rpm during an autorotation could not be determined. According to a Safety Notice published by the manufacturer, pilots with many hours of flight experience in airplanes and fewer hours experience in helicopters may instinctively make flight control inputs that are inappropriate for helicopters, which can result in catastrophic consequences.

Probable Cause and Findings

The National Transportation Safety Board determines the probable cause(s) of this accident to be:

The pilot’s failure to manage main rotor rpm during an autorotation, which resulted in a loss of helicopter control. Contributing to the accident was a loss of power for reasons that could not be determined.

Findings	
Aircraft	(general) - Unknown/Not determined
Aircraft	Prop/rotor parameters - Not attained/maintained
Personnel issues	Use of equip/system - Pilot

Factual Information

History of Flight

Enroute-cruise	Unknown or undetermined (Defining event)
Autorotation	Loss of control in flight
Landing	Hard landing

On June 19, 2024, about 1620 Pacific daylight time, a Robinson Helicopters R-44 II, N323TT, was destroyed when it was involved in an accident near Bluestem, Washington. The pilot and pilot-rated passenger were fatally injured. The helicopter was operated as a Title 14 *Code of Federal Regulations* Part 91 personal flight.

According to the owner, the helicopter was at a maintenance facility at Boeing Field/King County International Airport (BFI), Seattle, Washington, where it underwent an annual inspection and other maintenance. The owner asked the accident pilot to travel to BFI and return the helicopter to Coeur d’Alene/Pappy Boyington Field (COE), Coeur d’Alene, Idaho. The owner was also acquainted with the pilot-rated passenger but knew nothing about the details of the flight plan.

According to an acquaintance and trainee of the accident pilot, he flew the pilot and the pilot-rated passenger to BFI in his personal airplane. Once at BFI, he observed the pilot conduct a preflight inspection of the helicopter, and he saw a fuel truck top off the helicopter. The acquaintance departed BFI after the helicopter departed. While en route, he saw the helicopter and observed no indication of a problem. The acquaintance did not follow the same flight route as the helicopter and did not stop at Bowers Field (ELN), Ellensburg, Washington. He described the return flight weather as high clouds and no visibility restrictions, and the temperature was comfortable, but the ride was a little choppy.

FAA ADS-B data did not record the helicopter’s departure from BFI. ADS-B data, provided by a commercial source, showed that the helicopter departed BFI around 1338, climbed to a maximum altitude of about 2,000 ft mean sea level (msl), passed eastward over the Snoqualmie Pass, and proceeded directly to ELN. FAA ADS-B data first recorded the helicopter when it was about 10.5 miles northwest of ELN. The helicopter proceeded directly to the parking ramp, where it arrived about 1437. A fuel receipt from the ELN fixed-base operator recorded 14.38 gallons of 100LL fuel purchased for the helicopter. The helicopter departed ELN around 1511, climbed to a maximum altitude of about 3,927 ft msl, and proceeded northeast, consistent with flight toward COE. The helicopter generally maintained about 80 to 100 kts ground speed for most of the flight. The helicopter slowed momentarily when it neared Ephrata Municipal Airport (EPH), Ephrata, Washington, but accelerated back to about 80 to 100

kts after passing the airport. Although there were minor changes in heading and the ground speed varied, there were no indications of notable maneuvering during the 1 hour, 21-minute flight. ADS-B data revealed that, about one quarter mile southwest of the accident site, the helicopter began to decelerate from 85 kts to 55 kts at the last recorded data point, about 3,050 ft msl (745 ft agl) and about 40 ft west of the accident site. No witnesses came forward to provide first-hand accounts of the accident.

Pilot Information

Certificate:	Airline transport; Commercial	Age:	68, Male
Airplane Rating(s):	Single-engine land; Single-engine sea; Multi-engine land	Seat Occupied:	Right
Other Aircraft Rating(s):	Helicopter	Restraint Used:	4-point
Instrument Rating(s):	Airplane; Helicopter	Second Pilot Present:	Yes
Instructor Rating(s):	Airplane multi-engine; Airplane single-engine; Helicopter	Toxicology Performed:	Yes
Medical Certification:	Class 2 With waivers/limitations	Last FAA Medical Exam:	June 2, 2023
Occupational Pilot:	Yes	Last Flight Review or Equivalent:	
Flight Time:	21300 hours (Total, all aircraft), 300 hours (Total, this make and model)		

Pilot-rated passenger Information

Certificate:	Private	Age:	63, Male
Airplane Rating(s):	Single-engine land; Single-engine sea	Seat Occupied:	Left
Other Aircraft Rating(s):	None	Restraint Used:	4-point
Instrument Rating(s):	None	Second Pilot Present:	Yes
Instructor Rating(s):	None	Toxicology Performed:	Yes
Medical Certification:	Class 2 With waivers/limitations	Last FAA Medical Exam:	April 6, 2017
Occupational Pilot:	No	Last Flight Review or Equivalent:	
Flight Time:			

The pilot's personal logbooks were not available for review. The pilot reported on his last medical examination 21,300 total flight hours and 50 flight hours in the last 6 months.

According to the owner of Inland Helicopters, Spokane, Washington, the accident pilot was a contract flight instructor for his company. The employer estimated that the accident pilot accrued at least 250 to 350 hours of flight experience in Robinson helicopters and held the Special Federal Aviation Regulations number 73 (SFAR 73) endorsement for the R-44. The company owner reported that the accident pilot initially provided flight instruction training to

the helicopter owner as part of the pilot's employment with the company, but that the training developed into personal flight training that was no longer associated with his company.

Aircraft and Owner/Operator Information

Aircraft Make:	ROBINSON HELICOPTER COMPANY	Registration:	N323TT
Model/Series:	R44 II	Aircraft Category:	Helicopter
Year of Manufacture:	2023	Amateur Built:	
Airworthiness Certificate:	Normal	Serial Number:	14626
Landing Gear Type:	Skid	Seats:	4
Date/Type of Last Inspection:	June 4, 2024 Annual	Certified Max Gross Wt.:	2500 lbs
Time Since Last Inspection:		Engines:	1 Reciprocating
Airframe Total Time:	60 Hrs as of last inspection	Engine Manufacturer:	Lycoming
ELT:	C91A installed, not activated	Engine Model/Series:	IO-540-AE1A5
Registered Owner:	On file	Rated Power:	
Operator:	On file	Operating Certificate(s) Held:	None

According to the maintenance facility's director of maintenance, the helicopter was at their facility for an annual inspection and other maintenance that included replacement of the horizontal stabilizer. The replacement stabilizer had not arrived and was not replaced. A review of the maintenance records revealed that the maintenance included the installation of a left electronic ignition system fixed timing module and a right magneto. A JP Instruments (JPI) fuel flow indicating system and a JPI fuel flow transducer were installed. The fuel flow transducer was mounted between the throttle body and the fuel flow divider.

Meteorological Information and Flight Plan

Conditions at Accident Site:	Visual (VMC)	Condition of Light:	Day
Observation Facility, Elevation:	KSKA, 2461 ft msl	Distance from Accident Site:	21 Nautical Miles
Observation Time:	16:55 Local	Direction from Accident Site:	76°
Lowest Cloud Condition:	Clear	Visibility	10 miles
Lowest Ceiling:	None	Visibility (RVR):	
Wind Speed/Gusts:	6 knots / None	Turbulence Type Forecast/Actual:	None / None
Wind Direction:	30°	Turbulence Severity Forecast/Actual:	N/A / N/A
Altimeter Setting:	30.05 inches Hg	Temperature/Dew Point:	22°C / 2°C
Precipitation and Obscuration:	No Obscuration; No Precipitation		
Departure Point:	Seattle, WA (BFI)	Type of Flight Plan Filed:	None
Destination:	Cour d' Alene, ID (COE)	Type of Clearance:	None
Departure Time:	13:28 Local	Type of Airspace:	Class G

A weather study revealed that the synoptic conditions indicated no frontal boundaries in the immediate vicinity of the accident site and a weak pressure gradient, which resulted in light and variable winds over eastern Washington.

The closest weather observations to the accident site reported low instrument meteorological conditions (IMC) to IMC during the morning hours due to low ceilings and visibilities in fog before 0840, which cleared to VMC around the time of the accident. Conditions were reported as light and variable wind, visibility 10 statute miles or more, clear skies below 12,000 ft, with temperatures near 70°F.

A high-resolution rapid refresh (HRRR) numerical model wind profile depicted a near surface wind from 040° at 6 kts with little variation in direction or speed through 5,000 ft. Above 5,000 ft, the wind shifted to the west with slowly increasing wind speed with height. Between 8,600 and 11,200 ft, the sounding depicted a potential for light turbulence due to a change in wind direction and an increase in wind speed.

Wreckage and Impact Information

Crew Injuries:	1 Fatal	Aircraft Damage:	Destroyed
Passenger Injuries:	1 Fatal	Aircraft Fire:	On-ground
Ground Injuries:		Aircraft Explosion:	None
Total Injuries:	2 Fatal	Latitude, Longitude:	47.546733,-118.15535

The helicopter impacted open, hilly terrain and came to rest upright at the first point of impact, on a heading of about 086°, at an elevation of 2,182 ft msl. A postaccident fire destroyed most of the wreckage (see figure 1).



Figure 1. Main wreckage (Source: Robinson Helicopters).

The empennage, including the tail rotor gearbox, tail rotor assembly, and about two ft of the tailcone, separated from the helicopter and was about 338 ft southwest of the fuselage (see figure 2). Examination of the empennage revealed no indication of smoke or soot deposits.



Figure 2. Helicopter empennage, including the tail rotor gearbox and tail rotor assembly.

Postaccident examination of the main wreckage revealed the attached section of the tailcone exhibited an overall flattened deformity and an upward “scorpion” bending. Both main rotor blades exhibited a significant downward bending along with chordwise creases. One main rotor blade exhibited a color transfer on the leading edge that matched the color of the paint on the tailcone. The main rotor hub surface, adjacent to the main rotor blades, had small arc-shaped scuff marks through the paint, consistent with upward coning and contact of the blade grips with the hub. Each of the elastomeric teeter stops exhibited deep impact marks. One spindle tusk exhibited bending, consistent with excessive downward movement of the parent spindle and blade. The aft section of one of the main landing gear skids was bent upward to about 45°. The left-seat flight controls were installed.

Examination of the engine revealed multiple fractures of the engine case, along with extensive thermal damage. The engine cooling fan exhibited upward crushing with no rotational signatures on the non-impacted top section of the fan. The upper sheave exhibited deep score marks and dents, along with a buildup of material at the end of one deep score mark that showed movement in the direction opposite of normal rotation. Most of the external components of the engine were extensively thermally damaged. Spark plug examination revealed some exhibited normal color when compared with a Champion Check-A-Plug chart,

some exhibited impact damage, and at least one exhibited white color and some corrosion. All fuel injectors were unobstructed. The fuel servo was thermally damaged. All lines normally connected to the fuel servo were thermally damaged and separated, except for the line from the fuel flow transducer to the flow divider, which remained attached to the fuel servo. Multiple B-nuts from the fuel lines that were associated with the fuel flow transducer and fuel servo could be turned by hand. Thermal damage precluded further examination.

Medical and Pathological Information

An autopsy of the pilot was performed by the Spokane County Medical Examiner, Spokane, Washington. The cause of death was blunt force injuries.

Additional Information

Robinson Safety Notice SN-10, Fatal Accidents Caused by Low RPM Rotor Stall, issued in October 1982, and revised in February 1989 and June 1994, states:

A primary cause of fatal accidents in light helicopters is failure to maintain rotor rpm. To avoid this, every pilot must have his reflexes conditioned so he will instantly add throttle and lower collective to maintain rpm in any emergency...Power available from the engine is directly proportional to rpm. If the rpm drops 10%, there is 10% less power. With less power, the helicopter will start to settle, and if the collective is raised to stop it from settling, the rpm will be pulled down even lower, causing the ship to settle even faster. If the pilot not only fails to lower collective, but instead pulls up on the collective to keep the ship from going down, the rotor will stall almost immediately. When it stalls, the blades will either "blow back" and cut off the tailcone or it will just stop flying, allowing the helicopter to fall at an extreme rate. In either case, the resulting crash is likely to be fatal.

No matter what causes the low rotor rpm, the pilot must first roll on throttle and lower the collective simultaneously to recover rpm before investigating the problem. It must be a conditioned reflex. In forward flight, applying aft cyclic to bleed off airspeed will also help recover lost rpm.

Robinson Safety Notice SN-24 ("Low RPM Rotor Stall Can be Fatal"), issued in September 1986, and revised in June 1994, states:

When the rotor stalls, it does not do so symmetrically because any forward airspeed of the helicopter will produce a higher airflow on the advancing blade than on the retreating blade. This causes the retreating blade to stall first, allowing it to dive as it goes aft while the advancing blade is still climbing as it goes forward. The resulting low aft blade and high forward blade become a rapid aft tilting of the rotor disc sometimes referred to as "rotor blow-back." Also, as the helicopter begins to fall, the upward flow of air under the tail surfaces tends to pitch the aircraft nose-down. These two effects, combined with aft cyclic by the pilot attempting to keep the nose from dropping, will frequently allow the rotor blades to blow back and chop off the tailboom as the stalled helicopter falls.

Robinson Safety Notice SN-29 ("Airplane Pilots High Risk When Flying Helicopters"), issued in March 1993, and revised in June 1994, states:

There have been a number of fatal accidents involving experienced pilots who have many hours in airplanes but with limited experience flying helicopters.

The ingrained reactions of an experienced airplane pilot can be deadly when flying a helicopter. The airplane pilot may fly the helicopter well when doing normal maneuvers under ordinary conditions when there is time to think about the proper control response. But when required to react suddenly under unexpected circumstances, he may revert to his airplane reactions and commit a fatal error...For example, in an airplane his reaction to a warning horn (stall) would be to immediately go forward with the stick and add power. In a helicopter, application of forward stick when the pilot hears a horn (low rpm) would drive the rpm even lower and could result in rotor stall, especially if he also "adds power" (up collective).

The FAA Helicopter Flying Handbook (FAA-H-8083-21B) states:

The pilot should immediately apply simultaneous down collective, aft cyclic and trim the helicopter for entry into an autorotation initiated at cruise airspeed. If rotor rpm has been allowed to decrease, or has inadvertently decreased below acceptable limits, an application of aft cyclic may help rebuild rotor rpm. This application of aft cyclic must be made at least at a moderate rate and may be combined with a turn, either left or right, to increase airflow through the rotor

system. This will work to increase rotor rpm. Care should be maintained to not over-speed the rotor system as this is attempted.

Administrative Information

Investigator In Charge (IIC): Salazar, Fabian

Additional Participating Persons: Troy Helgeson; Lycoming Engines; Williams Port, PA
Todd Pryor; Federal Aviation Administration; Spokane, WA
Thom Webster; Robinson Helicopters; Torrence, CA

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Investigation Class: [Class 3](#)

Note:

Investigation Docket: <https://data.nts.gov/Docket?ProjectID=194501>

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